

-
-
- S1 ▾
- Q 1
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S1

Section A

Q1 >

Q2

Array Rotation

Flag Question

Problem Description

Given an integer array **A** of size **N** and an integer **B**, you have to return the same array after rotating it **B** times towards the right.

Problem Constraints

1 <= N <= 10⁵
1 <= A[i] <= 10⁹
1 <= B <= 10⁹

Input Format

The first argument given is the integer array A.
The second argument given is the integer B.

Output Format

Return the array A after rotating it B times to the right

Example Input

Input 1:

A = [1, 2, 3, 4]
B = 2

Input 2:

A = [2, 5, 6]
B = 1

Example Output

Output 1:

[3, 4, 1, 2]

Output 2:

[6, 2, 5]

Example Explanation

Explanation 1:

Rotate towards the right 2 times - [1, 2, 3, 4] => [4, 1, 2, 3] => [3, 4, 1, 2]

Explanation 2:

Rotate towards the right 1 time - [2, 5, 6] => [6, 2, 5]

You only need to implement the given function. Do not read input; instead use the arguments to the function. Do not print the output; instead return values as specified. Still have a question? Check out Sample Codes for more details.

- HINTS
- Video Explanation

New
- Hint 1
- Solution Approach
- Complete Solution

Editor Mode - Normal

Java 8 (oracle-jdk-1.8)

```
1 public class Solution {
2     public ArrayList<Integer> solve(ArrayList<Integer> A, int B) {
3         B= B%A.size();
4         ArrayList<Integer> ans = new ArrayList<>();
5         // for(int i = 0;i<A.size()-B;i++){
6             //     ans.add(A.get(i));
7             // }
8         // if(ans.size()==A.size()){
9             //     return ans;
10        // }else{
11        //     for(int i = A.size()-1;i>B+2;i--){
12            //         ans.add(0,A.get(i));
13            //     }
14            // }
15            // return ans;
16        //     for(int i = 0;i<=A.size()+B;i++){
17            //         if(i<=B){
18            //             ans.add(0);
19            //         }else{
20            //             ans.add(A.get(i-B));
21            //         }
22            //     }
23        //     for(int i = A.size()-1;i<A.size()+B;i++){
24        //         System.out.println(i);
25        //         ans.set(i-A.size()+1,ans.get(i));
26        //         ans.remove(i);
27        //         i--;
28        //     }
29        //     return ans;
30        //     if(B==0) return A;
31        //     int count = 0;
32        //     for(int i = 0;i<A.size()-B; i++){
33            //         ans.add(A.get(i));
34            //         count++;
35            //     }
36        //     for(int i= A.size()-1;i>=count;i--){
37        //         System.out.println(A.get(i));
38        //         ans.add(0,A.get(i));
39        //     }
40        //     return ans;
41    }
42 }
43 }
44 }
45 }
46 }
```

Reset

Test

Test with Custom Input

Submit

Correct Answer.

S1 ▾

Q 1

Q 2

S2 ▾

Q 3

Q 4

Q 5

Q 6

Q 7

Q 8

Q 9

Q 10

Q 11

Q 12

Q 13

S1 Section A

Q1

 >

Q2

Return vowels from every row

Flag Question

Problem Description

You are given a list of strings, A. Your task is to extract all the vowels (both uppercase and lowercase) from each string and return them as a 2D character array.

Each row of the 2D array corresponds to the vowels extracted from a single string in the input list, maintaining the order in which they appear.

If a string has no vowels, the corresponding row will be an empty list [].

Problem Constraints

Input will consist of at least one string.

Each string will contain only alphabets (A-Z, a-z) and spaces.

Vowels to be considered are: a, e, i, o, u (case insensitive).

Input Format

A list of strings.

Output Format

A 2D character array where each row contains the vowels extracted from the corresponding string in the input list.

If a string is empty or has no vowels, the corresponding row will be an empty list [].

Example Input

Input 1:

```
[
  "Java Programming",
  "Array Data Structure",
  "Computer"
]
```

Input 2:

```
[
  "Rhythm",
  "Python"
]
```

Example Output

Output 1:

```
[
  ['a', 'a', 'o', 'a', 'i'],
  ['A', 'a', 'a', 'a', 'a', 'u', 'u', 'e'],
  ['o', 'u', 'e']
]
```

Output 2:

```
[
  [],
  ['o']
]
```

Example Explanation

Explanation 1:

For "Java Programming", the vowels are a, a, o, a, i.

For "Array Data Structure", the vowels are A, a, a, a, a, u, u, e.

For "Computer", the vowels are o, u, e.

Explanation 2:

For "Rhythm", no vowels exist, so return [].

For "Python", the vowel is o.

HINTS

Hint 1

Solution Approach

Complete Solution

Editor Mode - Normal ▾ Java7 (open-jdk-1.7.0) ▾

- Hide Solution Prefix

```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;

1 class Solution {
2     public ArrayList<ArrayList<Character>> extractVowels(ArrayList<String> strings) {
3         // Your Code goes here
4         ArrayList<ArrayList<Character>> answer = new ArrayList<>();
5         for(int i = 0;i<strings.size();i++){
6             String string = strings.get(i);
7             ArrayList<Character> vowels = new ArrayList<>();
8             for(int j =0;j<string.length();j++){
9                 char c = string.charAt(j);
10                if(c=='a' ||c =='A' || c=='e' ||c=='E' ||c=='i' || c=='I' || c=='o' || c=='O' || c=='u' || c=='U'){
11                    vowels.add(c);
12                }
13            }
14            answer.add(vowels);
15        }
16        return answer;
17    }
18 }
19
20
```

+ Show Solution Suffix

Reset

Test Test with Custom Input

Submit

Correct Answer.



S1

Q 1

Q 2

S2

Q 3

Q 4

Q 5

Q 6

Q 7

Q 8

Q 9

Q 10

Q 11

Q 12

Q 13

S2 Section B

[Q3 >](#) [Q4 >](#) [Q5 >](#) [Q6 >](#) [Q7 >](#) [Q8 >](#) [Q9 >](#) [Q10 >](#) [Q11 >](#) [Q12 >](#) [Q13](#)

StringBuilder Manipulation

[Flag Question](#)

Fill in the blank with the correct output of this program:

```
String str = "abcd";
String rev = "";
for (int i = str.length() - 1; i >= 0; i--) {
    rev = rev + str.charAt(i);
}
rev = rev + "xy";
System.out.println(rev);
```

dcbaxy

 Your answer to this question has been recorded. It will be evaluated after the end of the contest. If you change your answer, kindly re-submit.

Re-Submit





S1

Q 1

Q 2

S2

Q 3

Q 4

Q 5

Q 6

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S2 Section B

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Function-1

[Flag Question](#)

Which of the following statements about functions (methods) in Java are correct?

 HINTS[Solution Approach](#)☒ Functions help in reusing code and making programs organized.☐ Functions must always return a number.☒ Functions can return a value or return nothing (void).☐ Functions can be defined inside other functions.

Expected Answer: Functions can return a value or return nothing (void), Functions help in reusing code and making programs organized.

 Your answer to this question has been recorded. It will be evaluated after the end of the contest. If you change your answer, kindly re-submit.



Re-Submit



CORRECT ANSWER

Your answer has been evaluated.





S1

Q 1

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Q 12

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Strings - 1

[Flag Question](#)

Which statements about Java String are correct?

 HINTS[Solution Approach](#)

- ☒ substring(begin, end) returns characters from index begin (inclusive) to index end (exclusive).
- ☒ trim() removes spaces only from the beginning and end of a string, not the middle.
- ☒ charAt(i) returns the character at position i in the string.
- ☒ contains("abc") returns true if the string contains the substring "abc"

Expected Answer: substring(begin, end) returns characters from index begin (inclusive) to index end (exclusive), trim() removes spaces only from the beginning and end of a string, not the middle, contains("abc") returns true if the string contains the substring "abc", charAt(i) returns the character at position i in the string.

 Your answer to this question has been recorded. It will be evaluated after the end of the contest. If you change your answer, kindly re-submit.



Re-Submit





S1

Q 1

Q 2

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Q 3

Q 4

Q 5

Q 6

Q 7

Q 8

Q 9

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S2 Section B

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String 2

[Flag Question](#)

What will be the output of the following code?

```
public class Main {  
    public static void main(String[] args) {  
        String s = "aabbcccddee";  
        System.out.println(hasTriplets(s));  
    }  
  
    public static boolean hasTriplets(String s){  
        int n = s.length();  
        for(int i=0; i<=n-3; i++){  
            if(s.charAt(i)==s.charAt(i+1)&& s.charAt(i)==s.charAt(i+2))  
                return true;  
        }  
        return false;  
    }  
}
```

 HINTS[Complete Solution](#)

true

 Your answer to this question has been recorded. It will be evaluated after the end of the contest. If you change your answer, kindly re-submit.

Re-Submit





S1

Q 1

Q 2

S2

Q 3

Q 4

Q 5

Q 6

Q 7

Q 8

Q 9

Q 10

Q 11

Q 12

Q 13

S2

Section B

Q3 >

Q4 >

Q5 >

Q6 >

Q7 >

Q8 >

Q9 >

Q10 >

Q11 >

Q12 >

Q13

Wave Print (Row Wise)

Flag Question

Problem Description

You are given an integer **N** and an **N x N** matrix.
Print all elements of the matrix in a row-wise wave traversal.

The traversal should look like this:

- Start with the first row, traverse from left to right.
- For the next row, traverse in the reverse direction of the previous row (so if the previous was left → right, now it's right → left).
- For the following row, switch the direction again.
- Continue alternating the direction for every subsequent row until the last row is covered.

Problem Constraints

1 <= N <= 10³

0 <= Mat[i][j] <= 10⁹

Input Format

First line is an integer **N**

Next **N** lines contain **N** space separated integers representing the matrix **Mat**

Output Format

Print all elements of the matrix in a single line, separated by spaces,
following the row-wise wave traversal order.

Note: Ensure there is a space character (' ') at the end of the line.

Example Input

Input 1:

```
3
4 1 2
7 4 4
2 7 4
```

Input 2:

```
2
1 2
3 4
```

Example Output

Output 1:

```
4 1 2 4 4 7 3 7 4
```

Output 2:

```
1 2 4 3
```

Example Explanation

For Input 1:

We will first iterate the 1st row from left to right and print the elements,
then we will iterate the 2nd row from right to left and print the elements,
then we will iterate the 3rd row from left to right and print the elements.
This looks like a wave.

For Input 2:

We will first iterate the 1st row from left to right and print the elements,
then we will iterate the 2nd row from right to left and print the elements.
This looks like a wave.

HINTS



Video Explanation

New

Hint 1

Solution Approach

Complete Solution



Editor Mode - Normal



Java 8 (oracle-jdk-1.8)



```
1 import java.lang.*;
2 import java.util.*;
3
4 public class Main {
5     public static void main(String[] args) {
6         // YOUR CODE GOES HERE
7         // Please take input and print output to standard input/output (stdin/stdout)
8         // DO NOT USE ARGUMENTS FOR INPUTS
9         // E.g. 'Scanner' for input & 'System.out' for output
10        Scanner sc = new Scanner(System.in);
11        int n = sc.nextInt();
12        ArrayList<ArrayList<Integer>> answer = new ArrayList<>();
13
14        for(int i = 0;i<n;i++){
15            ArrayList<Integer> row = new ArrayList<>();
16            for(int j =0;j<n;j++){
17                int temp = sc.nextInt();
18                row.add(temp);
19            }
20            answer.add(row);
21        }
22
23        boolean check = true;
24        for(int i = 0;i<n;i++){
25            if(check){
26                for(int j = 0;j<n;j++){
27                    System.out.print(answer.get(i).get(j)+" ");
28                }
29                check = false;
30            }else{
31                for(int j= n-1;j>=0;j--){
32                    System.out.print(answer.get(i).get(j)+" ");
33                    check = true;
34                }
35            }
36        }
37
38
39    }
40
41 }
```

Reset

Test

Test with Custom Input

Submit

Correct Answer.



S1

Q 1

Q 2

S2

Q 3

Q 4

Q 5

Q 6

Q 7

Q 8

Q 9

Q 10

Q 11

Q 12

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1D Arrays - 1

[Flag Question](#)

Which statements about Java 1D arrays are correct?

 HINTS[Solution Approach](#)

- ☒ Array elements are stored in contiguous memory locations.
- ☐ Array length is accessed by `arr.length()`.
- ☒ Arrays can store elements of only a single data type. For example, an `int[]` array can store only integers, not floats or strings.
- ☒ Arrays in Java are indexed from 0 to length-1.

Expected Answer: Arrays in Java are indexed from 0 to length-1.,Arrays can store elements of only a single data type. For example, an `int[]` array can store only integers, not floats or strings.,Array elements are stored in contiguous memory locations.

 Your answer to this question has been recorded. It will be evaluated after the end of the contest. If you change your answer, kindly re-submit.



Re-Submit



CORRECT ANSWER

Your answer has been evaluated.





S1

Q 1

Q 2

S2

Q 3

Q 4

Q 5

Q 6

Q 7

Q 8

Q 9

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Functions

[Flag Question](#)

Fill in the blank with the correct output of this program:

```
public class Main {  
    static String reverse(String s) {  
        String result = "";  
        for (int i = s.length() - 1; i >= 0; i--) {  
            if (i % 2 == 0) result += s.charAt(i);  
        }  
        return result;  
    }  
  
    public static void main(String[] args) {  
        System.out.print(reverse("abcdef"));  
    }  
}
```

eca

 Your answer to this question has been recorded. It will be evaluated after the end of the contest. If you change your answer, kindly re-submit.

Re-Submit



S1

Q 1

Q 2

S2

Q 3

Q 4

Q 5

Q 6

Q 7

Q 8

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Q 10

Q 11

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S2 Section B

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Q10 >

Q11 >

Q12 >

Q13

Remove all numbers with count of factors = K

Flag Question

Problem Description

Write a Java program (Oracle only) to input an **ArrayList<Integer> arr** and an integer **k**. For each number in **arr**, count its total number of factors (divisors).

Remove all numbers whose factor count is exactly **k** and return the remaining numbers in the same order using an **ArrayList**.
If no numbers remain, return an empty **ArrayList**.

You must solve this problem using Java (Oracle only) and ArrayList.

Problem Constraints

1 <= arr.size() <= 10³

1 <= arr.get(i) <= 10³

1 <= k <= 100

Input Format

The function/program takes the following inputs:

arr – ArrayList of integers
k – integer representing the factor count to remove

Output Format

Return an **ArrayList<Integer>** containing numbers from **arr** whose factor count is **not equal** to **k**.

The order of elements must be the same as in **arr**.
Return an empty list if no numbers remain.

Example Input

Input 1:

arr = [10, 15, 21, 7] k = 4

Input 2:

arr = [2, 3, 4, 5, 6] k = 2

Example Output

Output 1:

[7]

Output 2:

[4, 6]

Example Explanation

Explanation 1:

10 → factors = {1, 2, 5, 10} → 4 factors → removed
15 → factors = {1, 3, 5, 15} → 4 factors → removed
21 → factors = {1, 3, 7, 21} → 4 factors → removed
7 → factors = {1, 7} → 2 factors → kept
Final ArrayList = [7]

Explanation 2:

2 → 2 factors → removed
3 → 2 factors → removed
4 → 3 factors → kept
5 → 2 factors → removed
6 → 4 factors → kept
Final ArrayList = [4, 6]

HINTS

Hint 1

Solution Approach

Complete Solution

Editor Mode - Normal

Java 8 (oracle-jdk-1.8)

- Hide Solution Prefix

import java.util.*;

1

public class Main {

2

public static ArrayList<Integer> removeWithKFactors(ArrayList<Integer> arr, int k) {

3

// TODO: implement function

4

for(int i =0;i<arr.size();i++){

5

int num = arr.get(i);

6

int factors = 0;

7

for(int j = 1;j*j<=num;j++){

8

if(j*j==num){

9

factors++;

10

}else if(num%j==0){

11

factors+=2;

12

}

13

}

14

if(factors==k){

15

arr.remove(i);

16

i--;

17

}

18

}

19

return arr;

20

}

21

22

+ Show Solution Suffix

Reset

Test

Test with Custom Input

Submit

Correct Answer.



S1

Q 1

Q 2

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Q 4

Q 5

Q 6

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Q 9

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ArrayList - 1

[Flag Question](#)

Which statements about Java ArrayList are correct?

 HINTS[Solution Approach](#)

- ☒ ArrayList can change its size when we add or remove elements.
- ☐ Once created, the size of an ArrayList cannot be changed.
- ☐ An ArrayList has a fixed size that cannot be changed after it is created.
- ☒ We can use methods like add(), remove(), and get() with an ArrayList.

Expected Answer: ArrayList can change its size when we add or remove elements.,We can use methods like add(), remove(), and get() with an ArrayList.

 Your answer to this question has been recorded. It will be evaluated after the end of the contest. If you change your answer, kindly re-submit.

[Re-Submit](#)

CORRECT ANSWER

Your answer has been evaluated.





S1

Q 1

Q 2

S2

Q 3

Q 4

Q 5

Q 6

Q 7

Q 8

Q 9

Q 10

Q 11

Q 12

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S2 Section B

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2D Array Traversal

[Flag Question](#)

When the following code is executed, which of the following values of given 2D array will be printed?

```
int[][] mat = {  
    {1, 2, 3},  
    {4, 5, 6},  
    {7, 8, 9}  
};  
for (int i = 0; i < mat.length; i++) {  
    System.out.print(mat[i][mat.length - 1 - i] + " ");  
}
```

☒ 3☒ 7☒ 5☐ 9

☒ Your answer to this question has been recorded. It will be evaluated after the end of the contest. If you change your answer, kindly re-submit.



Re-Submit



S1

Q 1

Q 2

S2

Q 3

Q 4

Q 5

Q 6

Q 7

Q 8

Q 9

Q 10

Q 11

Q 12

Q 13

S2 Section B Q3 > Q4 > Q5 > Q6 > Q7 > Q8 > Q9 > Q10 > Q11 > Q12 > Q13

Solid Diamond - Without Space Flag Question

Problem Description

You are given an integer **N**. Print the corresponding Diamond Pattern consisting of **2 × N rows** without any extra space on right (for each row).

Problem Constraints

2 ≤ N ≤ 100

Input Format

The first and only line of input contains a single integer **N**.

Output Format

Print the diamond pattern corresponding to the given **N**. Ensure that there is no extra space after each '*' .

Example Input

Input 1: 2
Input 2: 3
Input 3: 5

Example Output

Output 1:

```
*
***
***
*
```

Output 2:

```
*
***
****
****
***
*
```

Output 3:

```
*
***
****
*****
*****
*****
*****
*****
****
***
*
```

Example Explanation

For each input value of N, print the diamond pattern as described. The first half increases the number of stars per row until N, and the second half decreases them symmetrically.

HINTS Hint 1 Solution Approach Complete Solution

Editor Mode - Normal Java 8 (oracle-jdk-1.8)

```
1 import java.lang.*;
2 import java.util.*;
3
4 public class Main {
5     public static void main(String[] args) {
6         // YOUR CODE GOES HERE
7         // Please take input and print output to standard input/output (stdin/stdout)
8         // DO NOT USE ARGUMENTS FOR INPUTS
9         // E.g. 'Scanner' for input & 'System.out' for output
10        Scanner sc = new Scanner(System.in);
11        int n = sc.nextInt();
12
13        for(int i = 0;i<n;i++){
14
15            for(int j =0;j<2*n;j++){
16                if(j<n-i-1){
17                    System.out.print(" ");
18                }else if(j>=n-i && j<=n+i){
19                    System.out.print("*");
20                }
21            }
22            System.out.println();
23        }
24        for(int i = n-1;i>=0;i--){
25            for(int j =0;j<2*n;j++){
26                if(j<n-1-i){
27                    System.out.print(" ");
28                }else if(j>=n-i && j<=n+i){
29                    System.out.print("*");
30                }
31            }
32            System.out.println();
33        }
34    }
35 }
36
37 }
```

Reset Test Test with Custom Input Submit